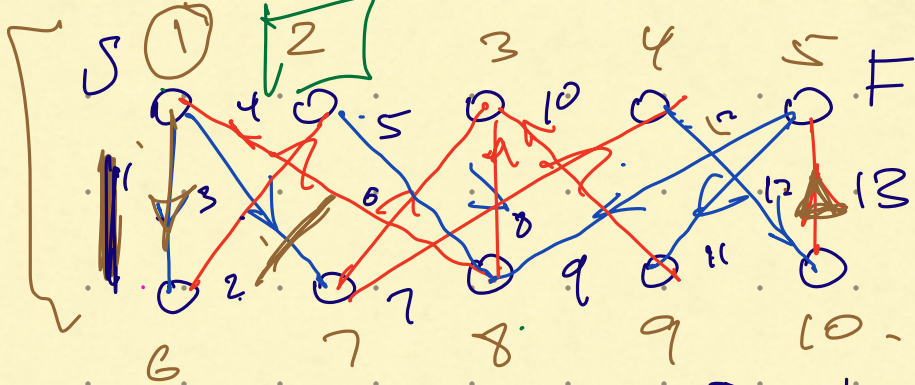
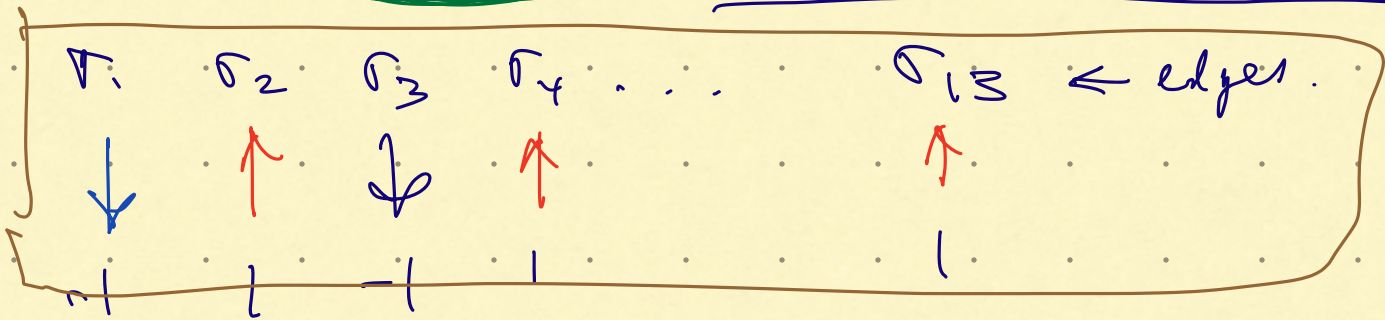




BFS DFS

Dijkstra algorithm



$r = 1, \dots, 10$ nodes 1

$$Q(r) = \sum_{\text{incident}(r)} \sigma_r \neq q_0(r) \cdot \sigma_0$$

$$\text{incident}(1) = \{1, 3, 4\}$$

$$\dots (3) = \{6, 8, 10\}$$

$$q_0(r) = \begin{cases} Q(r) = 0 & \text{if } r \text{ is not start or finish} \\ Q(r) = \pm 2 & \end{cases}$$

enclosed pass through

+2 if on top

-2 if on the bot

$$Q(1) = Q(5) = 2$$

enclosed start finish

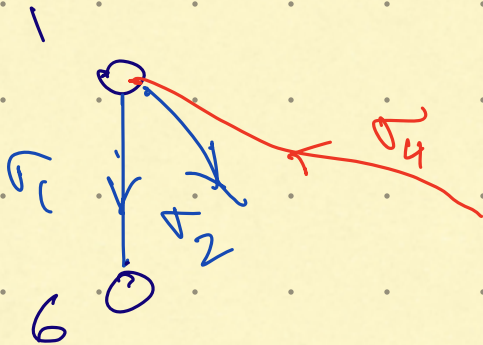
$$q_0(1) = 3 \quad q_0(2) = 0 \dots$$

Find such $\sigma'_1, \dots, \sigma'_{18}$

that $Q(r; \sigma') = 0$ for all r

then path is $\{\sigma_i, \sigma'_i\}$

if $\sigma_i, \sigma'_i = -1$ then edge i is in the path.



$$\sigma_i = -1 \rightarrow \sigma'_i = +1$$

$$\underline{Q(r=1; \sigma)} \rightarrow Q(r=1; \sigma')$$

$$= \underline{Q(r=1; \sigma) + 2}$$

$$\underline{Q(r=1; \sigma') = 0}$$

$$Q(r=6; \sigma') = 2$$



← flip both $\Rightarrow Q$ is unchanged.

Problem = representation.

Computing engine

$$Q(\sigma) = 0.$$

Constraint programming

(LLM)

Using machines, architectures,

$$Q^2(r) = \underline{\sigma}^1 = \arg\min_{\underline{\sigma}} Q^2(\underline{\sigma})$$

$$\underline{\sigma}^1 = \arg\min_{\underline{\sigma}} \sum_r Q^2(r; \underline{\sigma}).$$

$$Q^2(r; \underline{\sigma}) = \left(\sum_{m \in \text{inc}(r) \cup \{0\}} q_m(r) \sigma_m \right)^2$$

$$\underline{r=1}: m \in \{0, 1, 3, 4\}$$

$$q_0(r), q_{m \neq 0}(r) \approx 1$$

$$\left(\sum_m q_m(r) \sigma_m \right) \left(\sum_{m'} q_{m'}(r) \sigma_{m'} \right) =$$

$$= \sum_{m, m'} \underbrace{q_m(t)} \underbrace{q_{m'}(t)} \underbrace{\sigma_m \sigma_{m'}} = \underline{\underline{Q^2(t)}}$$

Ising model, $\mathcal{H}(\underline{\sigma}) = \frac{1}{2} \sum_{m, m'} \underline{\underline{A_{mm'}}} \sigma_m \sigma_{m'}$
Hamiltonian

interested in $\underline{\sigma}^* = \arg\min_{\underline{\sigma}} \mathcal{H}(\underline{\sigma})$

1) ground is (slightly) more hard,

2) No general algorithms,

3) It seems we have an engine
for $Q(\sigma) \sim 0$,

1) $\frac{1}{2}$ model of R-DLM.

2) Networks of oscillators
(Kuramoto model, Coherent Ising machine...)
ring oscillators $\rightarrow$

3) Probabilistic computing,

q-bit

4) Reservoir computing.

i) Can we recognize computing?